

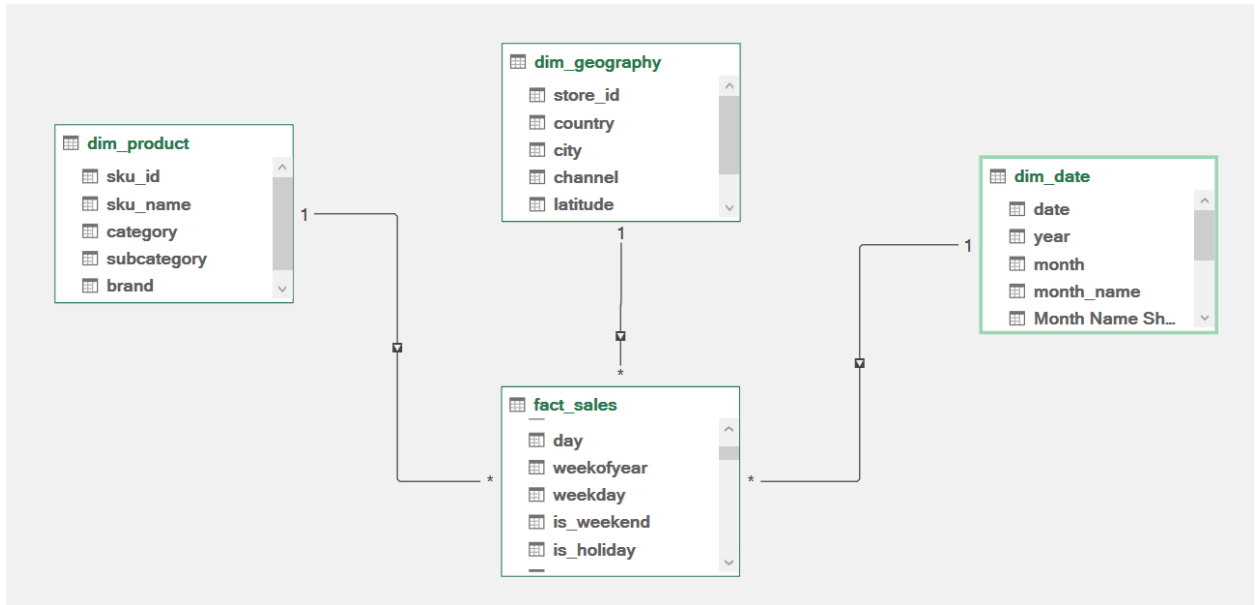
FMCG Commercial & Supply Chain Performance Documentation

Overview / Executive Summary

As a Data Analyst, I audited €472.95M in net sales to address promotional margin erosion and high inventory volatility across 5 FMCG categories in European markets. By engineering a dynamic analytical model, I identified that while promotions drive an 88.67% unit uplift, they compress overall profit margins by -14.39%, establishing actionable rebalancing strategies to protect €182.21M in net profit.

Data Sources & Cleaning

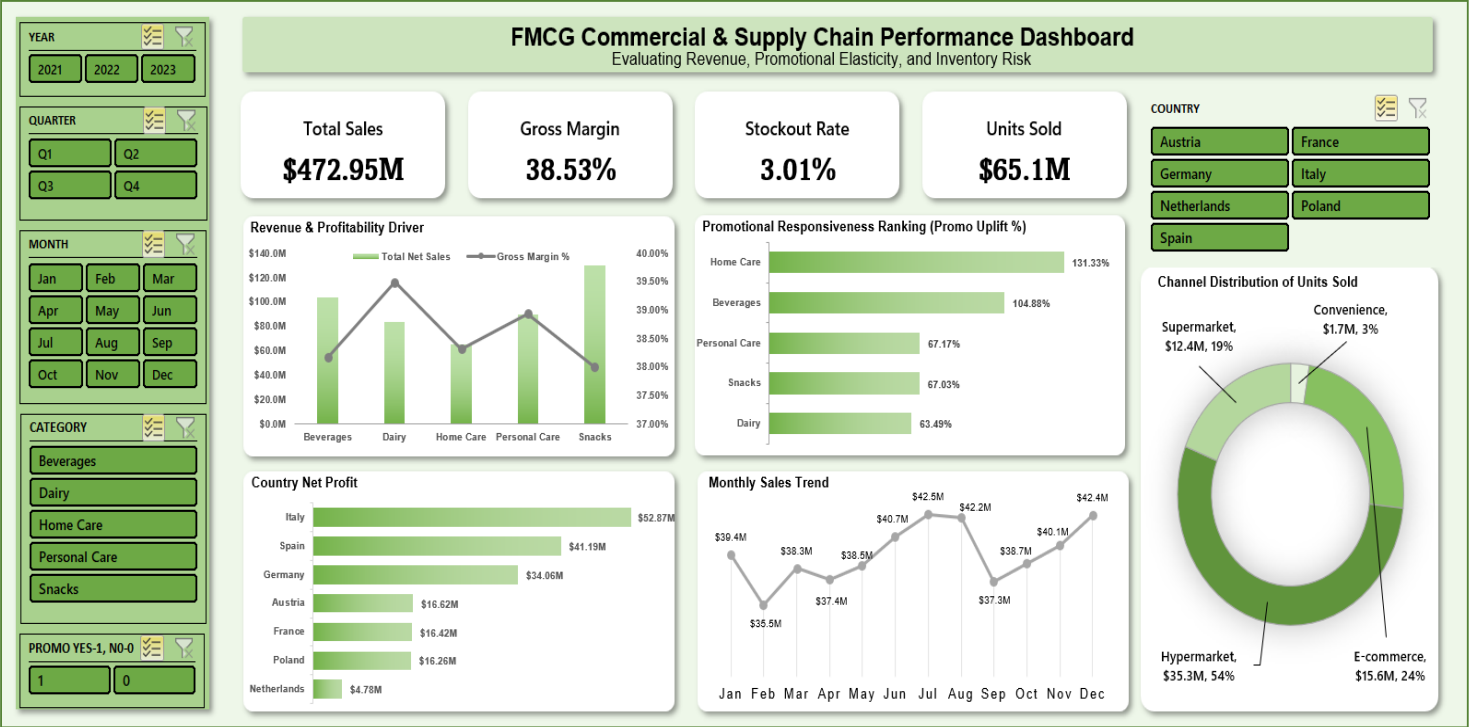
- **Source & Scope:** Extracted raw transactional data across 7 European countries, spanning **65.12M units sold** across Snacks, Beverages, Personal Care, Dairy, and Home Care categories.
- **Power Query ETL Pipeline:** Standardized text attributes (trimming whitespace and standardizing regional country names) and cast date strings into clean temporal hierarchies.
- **Data Quality Remediation:** Handled missing promotional flags using custom conditional columns in Power Query and removed duplicate order entries to ensure accurate revenue calculations.
- **Star Schema Architecture:** Separated raw flat data into distinct Fact and Dimension tables (Products, Geography, Promotion Types) linked via 1-to-many relationships in Power Pivot.



Methodology & Analysis

- **Advanced Dax & Measures:** Calculated key performance metrics, including Total Gross Revenue, Total Units Sold, Total Net Sales, Total Cost, Gross Margin, Promo Uplift, Average Discount, among others.
- **Inventory Volatility Modeling:** Applied standard deviation ($\sigma = 45.01$) and mean demand ($\mu = 59.20$) formulas to establish a **Coefficient of Variation ($CV = 0.76$)**, standardizing all categories under **XYZ Class (Z - Volatile)**.
- **Promotional Elasticity & Margin Spread:** Used dynamic PivotTables to measure unit uplift alongside margin compression across promotion and non-promotion periods.
- **Dynamic Excel Features:** Connected slicers across dynamic PivotCharts and KPI cards, using explicit number formatting to display monetary metrics cleanly.

Dashboard



Portfolio Summary

Category	Total Net Sales	Demand Volatility CV	XYZ Class	Promo Uplift %	Promo Margin Spread	Stockout Rate	Gross Margin %
Snacks	\$130.14M	0.7176	Z - Volatile	67.03%	-0.1433	3.04%	38.01%
Beverages	\$104.10M	0.7905	Z - Volatile	104.88%	-0.1429	3.03%	38.17%
Personal Care	\$89.55M	0.7719	Z - Volatile	67.17%	-0.1472	3.00%	38.94%
Dairy	\$83.82M	0.7115	Z - Volatile	63.49%	-0.1388	2.97%	39.50%
Home Care	\$65.33M	0.7044	Z - Volatile	131.33%	-0.1457	3.00%	38.32%
Grand Total	\$472.95M	0.7603	Z - Volatile	88.67%	-0.1439	3.01%	38.53%

Country	Total Net Sales	Gross Margin %	Promo Uplift %	Stockout Rate %
Netherlands	\$12.52M	38.20%	114.60%	3.04%
Poland	\$42.39M	38.36%	81.51%	2.96%
France	\$42.60M	38.55%	89.74%	3.07%
Austria	\$43.02M	38.64%	85.33%	3.09%
Germany	\$88.53M	38.48%	88.48%	3.01%
Spain	\$106.70M	38.60%	90.72%	2.99%
Italy	\$137.19M	38.53%	88.33%	3.00%
Grand Total	\$472.95M	38.53%	88.67%	3.01%

Baseline Calculations	
Total Net Sales	\$472.95M
Gross Margin %	38.53%
Stockout Rate %	3.01%
Total Units Sold	\$65.1M
Total Gross Revenue	\$484.75M
Total Cost	\$290.74M
Total Net Profit	\$182.21M
Promo Uplift %	88.67%
Avg Discount %	1.50%
Stockout Days	33114
Demand Standard Deviation	45.0072
Demand Mean	59.1963
Demand Volatility CV	0.7603
Recommended Safety Stock	189
Product Rev Share	1
Promo Sales Share %	10.44%
Promo Margin Spread	-0.143879063

Key Findings & Recommendations

- **Restructure High-Uplift Promotional Discounts:** Home Care (**131.33% uplift**) and Beverages (**104.88% uplift**) drive volume during campaigns but experience severe margin erosion (**-14.57%** and **-14.29%** margin spread, respectively). Transition these categories from deep price cuts to bundle offers or volume-tiering to capture revenue without sacrificing gross margin.

- **Buffer Safety Stock for Volatile Demands:** All 5 categories demonstrate high demand volatility ($CV > 0.70$). Implement a baseline safety stock target of **189 units per SKU** across distribution hubs to mitigate the **3.01% stockout rate** (~33,114 total stockout days) without over-leveraging working capital.
- **Scale High-Margin Anchor Categories:** Dairy delivers the highest gross margin (**39.50%**) despite lower promo sensitivity (**63.49% uplift**). Reallocate a portion of promotional spending from low-margin snacks to targeted marketing for Dairy products to improve net margins.
- **Target Regional Growth Markets:** Netherlands exhibits the strongest promotional response (**114.60% uplift**) despite representing the smallest sales footprint (**€12.52M**). Increase inventory allocations to the Netherlands during promotional periods to capture unmet demand.